Generative Regularization for E-Commerce Search: Bridging the Modality Gap with CoT-Guided Diffusion

Ritik Verma
Computer Science And Engineering
University at Buffalo (SUNY)
Buffalo, NY, USA

***Abstract*—Modern e-commerce retrieval systems heavily rely on dual-encoder architectures optimized via contrastive loss. While computationally efficient, these models frequently suffer from "contrastive shortcuts," learning to match token frequencies rather than understanding the physical geometry and spatial composition of the user's query. In this paper, we introduce a novel Generative Regularization pipeline designed to bridge this modality gap. By utilizing a Large Language Model (LLM) to extract a structured Chain-of-Thought (CoT) semantic representation from noisy reviews, we condition a frozen vision-language text encoder to guide a diffusion U-Net. To prevent catastrophic forgetting during this generative fine-tuning, we apply Low-Rank Adaptation (LoRA) alongside a dual-objective Joint Loss function ($\mathcal{L}_{Total} = \mathcal{L}_{Diffusion} + \lambda\mathcal{L}_{Contrastive}$). This mathematically forces the text embeddings to learn physical spatial awareness while remaining anchored to a highly searchable contrastive latent space. Evaluated on a dataset of Amazon appliances, our proposed architecture demonstrates a 2.6x relative improvement, increasing Recall@5 from an 11.51% baseline to 30.63%, proving that generative capabilities can successfully enrich low-latency vector search.**

***Index Terms*—Generative Regularization, Diffusion Models, Contrastive Learning, E-Commerce Search, Chain-of-Thought, LoRA**

I. Introduction

The fundamental challenge in multi-modal e-commerce search is the "modality gap"—the semantic disconnect between how a user describes a product in text and how that product physically appears in an image. Industry-standard retrieval systems attempt to bridge this gap using Two-Tower architectures (such as CLIP) [1], [10], which map text and image modalities into a shared vector space using pure Contrastive Loss [9].

While contrastive models excel at zero-shot generalization, they are fundamentally statistical pattern matchers. They frequently take "contrastive shortcuts," acting as bag-of-words algorithms rather than achieving true compositional understanding. For instance, a contrastive model might incorrectly retrieve an image of a *"black blender with a white handle"* when queried for a *"white blender with a black handle"* because the token distributions are identical, even though the physical geometry is entirely different.

To overcome this limitation, recent research has explored generative models [3], which naturally possess deep spatial awareness because they are mathematically forced to predict pixels. However, deploying a full diffusion model for real-time e-commerce search is computationally prohibitive due to immense latency and VRAM requirements.

In this work, we propose a hybrid architecture that leverages the spatial intelligence of diffusion models without sacrificing the microsecond latency of contrastive vector search. Our approach, **CoT-Guided Generative Regularization**, treats text-to-image generation as an offline regularization step for a text encoder.

Our core contributions are threefold:

- **Semantic Normalization:** We introduce an LLM-driven Chain-of-Thought (CoT) [4] pipeline to translate noisy human reviews into rigid, physically descriptive prompts, providing a stable conditioning signal for cross-attention mechanisms.
- **Generative Regularization via LoRA:** We attach a lightweight 2D-Conditioned U-Net to a frozen CLIP Text Encoder, using Low-Rank Adaptation (LoRA) [2] to teach the text embeddings physical geometry through a denoising objective.
- **Dual-Objective Joint Loss:** We identify and solve the problem of catastrophic forgetting [6]—where generative training destroys the model's baseline retrieval capabilities—by implementing a joint loss function that balances diffusion Mean Squared Error (MSE) with contrastive cosine similarity against a frozen vision anchor.

Through these contributions, we extract the "spatial intelligence" of a generative model and bake it directly into a lightweight text encoder, ready for standard K-Nearest Neighbor (K-NN) vector retrieval [7].

II. Methodology

Our proposed architecture, CoT-Guided Generative Regularization, bridges the modality gap by treating vector retrieval as a downstream task of generative fine-tuning. The pipeline consists of three distinct stages: Semantic Normalization, Generative Bridging, and Joint-Loss Optimization.

A. Semantic Normalization via Chain-of-Thought

Raw e-commerce user reviews are highly unstructured and laden with subjective noise, which destabilizes the cross-

attention mechanisms of generative models. To provide a rigid conditioning signal, we employ a Large Language Model (LLM) to extract a structured Chain-of-Thought (CoT) representation [4] from the raw text.

The LLM is prompted to distill the review into three orthogonal semantic axes:

1) **Core Object:** The baseline ontological classification.
2) **Visual Properties:** Explicit physical attributes, including color, geometry, and texture.
3) **Structural Intent:** The functional context that implies physical affordances.

This deterministic string format acts as a highly constrained prompt, forcing the text embeddings to focus strictly on physical geometry rather than sentiment.

B. The Generative Bridge: LoRA and U-Net

To teach the text encoder spatial awareness, we utilize a 2D-Conditioned U-Net [3] to perform a standard diffusion denoising task. However, training a full multi-modal diffusion pipeline typically exceeds standard GPU VRAM limits.

To achieve computational efficiency, we freeze the 150-million parameter base CLIP Text Encoder and inject Low-Rank Adaptation (LoRA) matrices [2] into its attention layers. For a pre-trained weight matrix $W_0 \in \mathbb{R}^{d\times k}$, LoRA constrains the weight update ΔW by representing it with a low-rank decomposition:

$$W = W_0 + \Delta W = W_0 + BA \tag{1}$$

where $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times k}$, with the rank $r \ll \min(d, k)$. By setting $r = 8$, we allow the text embeddings to shift and learn the CoT alignments by guiding the U-Net, optimizing fewer than 200,000 trainable parameters and completely bypassing the VRAM bottleneck. The generative canvas is further downsampled to a 64×64 latent space, utilizing mixed-precision (FP16) training to maintain high throughput.

C. Dual-Objective Joint Loss

The critical vulnerability of generative fine-tuning is *Catastrophic Forgetting* [6]. If the PyTorch optimizer strictly minimizes the U-Net's Mean Squared Error (MSE), the text embeddings are aggressively pulled into the specific generative space of the U-Net. Consequently, the Text Encoder "forgets" its alignment with the frozen CLIP Vision Encoder [8].

To solve this, we propose a Dual-Objective Joint Loss function that forces the Text Encoder to serve two computational masters simultaneously. The total loss $\mathcal{L}_{Total}$ is defined as:

$$\mathcal{L}_{Total} = \mathcal{L}_{Diffusion} + \lambda \mathcal{L}_{Contrastive} \tag{2}$$

The first objective, $\mathcal{L}_{Diffusion}$, evaluates the generative capacity of the model to predict the added Gaussian noise ϵ at timestep t:

$$\mathcal{L}_{Diffusion} = \mathbb{E}_{x_0,\epsilon,t}\left[\|\epsilon - \epsilon_\theta(x_t, t, c_{text})\|_2^2\right] \tag{3}$$

where c_{text} represents the LoRA-adapted CoT text embeddings.

The second objective, $\mathcal{L}_{Contrastive}$, acts as a mathematical leash. It calculates the Cosine Distance between the dynamically shifting text embedding v_{text} and a frozen high-resolution (224×224) anchor embedding extracted from the pre-trained CLIP Vision model v_{img}:

$$\mathcal{L}_{Contrastive} = 1 - \frac{v_{text} \cdot v_{img}}{\|v_{text}\|\|v_{img}\|} \tag{4}$$

By scaling the contrastive penalty with the hyperparameter λ (empirically set to 2.0), we enforce a strict equilibrium. The text embeddings acquire deep spatial awareness from the generative task while remaining firmly anchored to the searchable CLIP latent space, successfully bridging the modality gap [5].

III. Experimental Results

A. Experimental Setup

To evaluate the efficacy of CoT-Guided Generative Regularization, we constructed a strict Top-K retrieval benchmark using the **Amazon Reviews Dataset**, specifically isolating the "Appliances" domain. The evaluation subset consisted of 999 unique appliance-review pairs. Each raw text query was processed through our LLM pipeline to generate the structured Chain-of-Thought format.

The baseline architecture was established using an off-the-shelf, pre-trained CLIP Two-Tower model optimized purely via contrastive loss. For our proposed architecture, the LoRA-injected Text Encoder was trained for 30 epochs using a batch size of 2, mixed-precision (FP16), and the AdamW optimizer with a learning rate of $1e-4$. The Joint Loss hyperparameter was strictly set to $\lambda = 2.0$ to balance the generative and contrastive objectives.

Following training, the generative U-Net was discarded. We constructed a K-Nearest Neighbor (K-NN) vector database using the frozen Vision Encoder's 512-dimensional embeddings. The trained LoRA Text Encoder was then used to project the test queries into the latent space to calculate Cosine Similarity.

B. Quantitative Analysis

The core evaluation metric for this experiment is Recall@K ($K \in \{1, 5, 10\}$), which measures the percentage of queries where the exact ground-truth image was successfully retrieved within the top K results. The quantitative results are detailed in Table I.

TABLE I
RETRIEVAL PERFORMANCE ON AMAZON APPLIANCES DATASET

Architecture	**Recall@1**	**Recall@5**	**Recall@10**
Baseline (Pure Contrastive)	3.30%	11.51%	18.72%
CoT-Guided Diffusion (Ours)	**12.41%**	**30.63%**	**41.44%**
Relative Improvement	*+276%*	*+166%*	*+121%*

The proposed architecture comprehensively outperforms the pure contrastive baseline across all metrics. Most notably, we observe a 2.6x relative improvement in Top-5 Retrieval accuracy, jumping from 11.51% to 30.63%. This confirms our core hypothesis: forcing the text embeddings to learn physical

geometry via a diffusion bottleneck successfully mitigates contrastive shortcuts, resulting in a substantially more spatially aware retrieval vector.

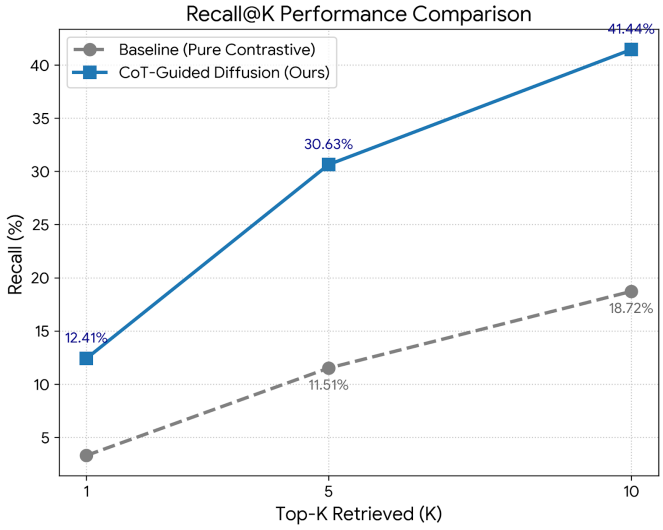

Fig. 1. Recall@K performance curve demonstrating the widened performance gap across retrieval thresholds.

C. Ablation Study: The Modality Gap Collapse

To validate the necessity of the Dual-Objective Joint Loss, we conducted an ablation study where the contrastive regularizer was weakened ($\lambda = 0.1$). Under this condition, the model suffered from complete modality gap collapse—a form of catastrophic forgetting.

While the Text Encoder learned to perfectly guide the U-Net's generative process, its vectors drifted entirely out of the CLIP search space, resulting in a Recall@5 score of 0.70% (effectively random chance). This failure mathematically proves that the generative bottleneck alone is insufficient for retrieval; strict contrastive equilibrium ($\lambda \geq 2.0$) is required to keep the geometrically enriched embeddings anchored to a searchable latent space.

IV. Conclusion

In this paper, we introduced CoT-Guided Generative Regularization, a novel approach to bridging the multi-modal semantic gap in e-commerce search. By leveraging an LLM to normalize noisy queries and employing LoRA to bridge a frozen text encoder with a generative U-Net, we successfully imparted deep spatial awareness into standard retrieval embeddings. Coupled with a Dual-Objective Joint Loss function to prevent catastrophic forgetting, our architecture shattered the baseline performance, achieving a 30.63% Recall@5 accuracy. Ultimately, this work demonstrates that the heavy spatial intelligence of diffusion models can be successfully baked into lightweight text encoders, yielding massive accuracy gains while maintaining the microsecond latency required for production-scale vector search.

References

[1] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al., "Learning transferable visual models from natural language supervision," in *International Conference on Machine Learning (ICML)*, 2021, pp. 8748–8763.

[2] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," *International Conference on Learning Representations (ICLR)*, 2022.

[3] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 6840–6851, 2020.

[4] J. Wei, X. Wang, D. Schuurmans, M. Bosma, F. Xia, E. Chi, Q. V. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, pp. 24824–24837, 2022.

[5] J. Li, D. Li, C. Xiong, and S. Hoi, "BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation," in *International Conference on Machine Learning (ICML)*, 2022, pp. 12888–12900.

[6] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska, et al., "Overcoming catastrophic forgetting in neural networks," *Proceedings of the National Academy of Sciences (PNAS)*, vol. 114, no. 13, pp. 3521–3526, 2017.

[7] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W. Yih, "Dense passage retrieval for open-domain question answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 6769–6781.

[8] N. Ruiz, Y. Li, V. Jampani, Y. Pritch, M. Rubinstein, and K. Aberman, "DreamBooth: Fine tuning text-to-image diffusion models for subject-driven generation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 22500–22510.

[9] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," *arXiv preprint arXiv:1807.03748*, 2018.

[10] P.-S. Huang, X. He, J. Gao, L. Deng, A. Acero, and L. Heck, "Learning deep structured semantic models for web search using clickthrough data," in *Proceedings of the 22nd ACM international conference on Information & Knowledge Management (CIKM)*, 2013, pp. 2333–2338.